



SAS Firmware Update Instructions

1200.2 SSD firmware update version 0004



This firmware update is for the Seagate 1200.2 SSD family only.

Backup all data contained on the drive prior to running the update.

See the End User License Agreement below for specific information about these software tools.

This firmware release only supports the following models:

1200.2 SSD (Koho). Select the appropriate firmware file for your drive.

Standard drives

Models	PN	Type
ST200FM0133	1XM2J2	High Endurance
ST400FM0233	1NH2J2	Mainstream Endurance
ST400FM0303	1NU252	Light Endurance
ST400FM0323	1XM2E2	High Endurance
ST480FM0003	1WX2J2	Light Endurance
ST800FM0173	1NH2K2	Mainstream Endurance
ST800FM0233	1NU2P2	Light Endurance
ST960FM0003	1WX2E2	Light Endurance
ST1600FM0003	1NH2L2	Mainstream Endurance
ST1600FM0073	1WX2R2	Light Endurance
ST1920FM0003	1NJ2F2	Light Endurance
ST1920FM0043	1YH272	Scalable Endurance

ST3200FM0023	1NH2M2	Mainstream Endurance
ST3200FM0063	1WX2S2	Light Endurance
ST3840FM0003	1NJ2G2	Light Endurance
ST3840FM0043	1YH282	Scalable Endurance

For the STD models above use firmware file [KohoSSD-STD-0004.LOD](#)

Self-Encrypting drives

Models	PN	Type
ST200FM0143	1Y12J2	High Endurance
ST400FM0243	1NT2J2	Mainstream Endurance
ST400FM0333	1Y12E2	High Endurance
ST400FM0343	1XR252	Light Endurance
ST480FM0013	1XR2J2	Light Endurance
ST800FM0183	1NT2K2	Mainstream Endurance
ST800FM0243	1XR2P2	Light Endurance
ST960FM0013	1XR2E2	Light Endurance
ST1600FM0013	1NT2L2	Mainstream Endurance
ST1600FM0083	1XS2R2	Light Endurance
ST1920FM0023	1XS2F2	Light Endurance
ST1920FM0053	1YJ272	Scalable Endurance
ST3200FM0033	1NT2M2	Mainstream Endurance
ST3200FM0073	1XS2S2	Light Endurance
ST3840FM0023	1XS2G2	Light Endurance
ST3840FM0053	1YJ282	Scalable Endurance

For the SED models above use firmware file [KohoSSD-SED-0004.LOD](#)

SED FIPS 140-2 Models (Review Pending)

Models	PN	Type
ST400FM0293	1VW2J2	Mainstream Endurance
ST800FM0213	1VW2K2	Mainstream Endurance
ST1600FM0023	1VW2L2	Mainstream Endurance
ST3200FM0043	1VW2M2	Mainstream Endurance

For the FIPS SED model above use firmware file [KohoSSD-FIPS-0004.LOD](#)

STD Standard
 SED Self Encrypting Drive
 FIPS Federal Information Processing Standard

If your current **1200.2 SSD** firmware does not end with the digits **0003**, then you may have a customized OEM configuration which is not compatible with **0004**. Please contact your OEM for firmware updates for these drives.

Warning, possible loss of data if this firmware is downloaded to unsupported models!

DO NOT run this firmware update on RAID systems.
DO NOT turn the power off during the firmware update procedure.

BEFORE YOU BEGIN:

1. Make sure you have backed up all of your important files and critical data.
2. Save any work in progress.
3. Close all other open applications.
4. Disconnect all external storage devices.

Always handle the drive carefully so you do not cause internal damage to the heads and media inside the drives. Drives are sensitive devices.

A warning about RAID

RAID systems are extremely sensitive to disruptions to individual drives. It is not uncommon for low level disk drive diagnostics to cause RAID management software to fault a drive that is slowed down by testing or firmware download. For this reason, we highly recommend that you use disk management tools provided by your RAID controller manufacturer if they are available. If these tools are not available, you might consider taking the RAID offline to a maintenance mode or rebooting the system to temporary bootable media and running diagnostics from there. Seagate recommends that you use disk utility tools provided by your RAID controller manufacturer.

Among others, RAID management software is available from 3Ware, Adaptec and LSI.

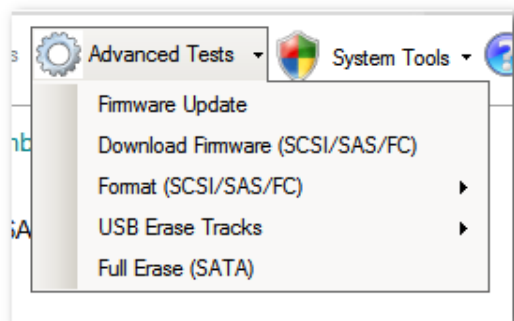
Here are the instructions for updating SAS firmware:

The method you choose to update the firmware on the drive will depend on your operating system, system utilities or work bench environment.

Windows Tool

Windows operating systems tolerate firmware downloads to SAS disk drives in JBOD mode. Drives in RAID configurations should not be given new firmware unless a special tool is provided by the RAID controller manufacturer.

Seagate's SeaTools for Windows may be used to download firmware to SAS disk drives in JBOD mode. You will find SeaTools for Windows at www.seagate.com/support/seatools and select the Downloads tab. After SeaTools is installed on the system, you will find *Download Firmware (SCSI/SAS/FC)* on the Advanced Tests menu.



The firmware files will be either .LOD or .SEA file types.

Important: These files must be copied to the SeaTools for Windows folder.

Linux Command Line Interface Tools

A command line interface tool is available for Linux based systems. **seafflashlin** firmware loader supports compatible Seagate disk drives listed on /dev/sg devices. The command line instructions for this tool are:

```
=====
Seagate Firmware Download Utility.
Copyright (c) 2016 Seagate Technology LLC, All Rights Reserved
SeaFlashLin Version: 0.4.7
=====

Usage
=====
    seafflashlin -f <fw_file> -d <sg_device>
    seafflashlin {-m <model_number> -f <fw_file> | -c <cfs_file>} [-d <sg_device>]

Example
=====
Specific drive:
    seafflashlin -f test.lod -d /dev/sg0
    (defaults to segment size = 64 and timeout = 5 seconds)

Multiple drives:
    seafflashlin -c barracuda.cfs
    seafflashlin -m ST600MP0005

Control options
=====
    -i          Display identification information (list sg_devices)

    -d, --device deviceHandle
                Use this option to specify the sg device handle on which to
                perform an operation. Multiple device handles can be given.

                Example: -d /dev/sg5

    -f <fw_file>
                The name of the firmware file.

    -x <path>
                To designate alternate location of the firmware files. Default
                is the current working directory.

    -c <cfs_file> (SATA only)
                A Seagate-prepared compatible firmware selector file. All
                drives in the system will be reviewed for new firmware based on
                this CFS file.
```

```
-k <cfs_file>    (SATA only)
    "dry run" test to determine which files and drives are
    supported by a given CFS file.

-m <model_number>
    The model number of the drive to download firmware. All drives
    is the system with this model number will be selected. Use a
    CFS first if you have one instead of using this choice.

    Not recommended for batch updates to multiple SATA notebook or
    desktop drives because the model numbers are not exclusively
    defined for just one family. In other words, two similar
    drives with the same model number may have different part
    numbers and will not share the same firmware file.
```

Modes and Controls

=====

```
-e    Deferred download. Not common. (ATA only)
-s    Segment size (1 segment = 512 bytes) Default is 64 (32KiB)
-p    Force ATA passthrough (for SATA)
-w    Force Write Buffer CMD (for SAS)
-u    Non-segmented
-o    Timeout value (seconds). Default without this option is 5
    seconds.
```

Utility Options

=====

```
-v    Display version and exit (overrides all other arguments)
-l    Show license agreement
```

Return codes

=====

```
0      No error found
1      Fatal error in command line options
2      FW Download Failed
3      Invalid device
4      No match in CFS file (SATA only)
5      FW up to date (SATA only)
6      Model number mismatch
7      FW mismatch
4x     Hexadecimal number count of Seagate sg devices in the
    system. This response code is only returned by the -i
    system scan argument. Beginning with 41 where x is the
    beginning of the count of the number of devices.
Anything else = unknown error
```

See the seaflashlin.txt file for a complete description of the software.

First, run the `-i` option to determine what `/dev/sg#` assignment lines up to your disk drive. This option also will show you other details about the drive including the current firmware revision. Like this:

```
/dev/sg0  MN: ST94813AS      SN: 3EE043KP  FW: 3.03
/dev/sg1  MN: ST1000NM0011   SN: Z1M15VAS  FW: SN03
```

`-m` is a model number verification based on the device Inquiry response. `-m` mode scans all sg devices on the system, so it is -- in itself -- a batch mode. The `-d` switch may be used with `-m` to select a specific device.

The raw tool has two batch modes, which can be used thusly:

```
seaflashlin -m <model_number> -f <fw_file>
```

Note: Use of -m mode is preferred for batch processing of sg (SCSI generic) SAS disk drives. The -m mode may also be used on Enterprise SATA disk drives. Desktop and notebook SATA drives firmware is usually combined with a Seagate-prepared CFS firmware configuration file.

The <cfs_file> noted above is a Seagate-prepared compatible firmware selector file which is used to match up the right firmware for a particular SATA drive.

```
seaflashlin -c <cfs_file>
```

Note: Batch processing of SATA drives should use a Seagate prepared CFS file.

Both of these modes assume that the relevant firmware files are located in the current working directory.

Alternately, these two batch modes may be targeted to a specific device like so:

```
seaflashlin -m <model_number> -f <fw_file> -d </dev/sg#>
```

```
seaflashlin -c <cfs_file> -d </dev/sg#>
```

By specifying -f and -d, without the benefit of requiring a model match with -m or utilizing a Seagate .CFS file, a user may still force a SATA download attempt on any /dev/sg device regardless of type or suitability. Obviously, this use is not recommended for most situations, but is available if needed. Multiple devices can be defined like this:

```
seaflashlin -f <fw_file> -d </dev/sg#> -d </dev/sg#> -d </dev/sg#>
```

Raw Firmware Data Files

You will need the Seagate firmware file to run the simple command line Linux tool. This file will have the .LOD filename extension and should be located in the current working directory. This firmware file should never be used outside of the specific product family models and part numbers listed with the firmware release. Incorrect usage of the .LOD may result in data loss or even product failure. Seagate Technology is not responsible for lost user data. See the End User License Agreement below for specific information about these software tools.

Using Traditional Linux Firmware Tools

For SAS drive firmware downloads you might use the Linux "sg_write_buffer" command. The sg command stands for the (SCSI Generic driver) and is an off-the-shelf Linux command that is kept updated by the Linux community. It may be able to

work with various RAID cards and see the individual drives. You will need to load the "sg3_utils" package if not already in your Linux kernel.

For Ubuntu `_ apt-get sg3_utils`

For Fedora/redhat/centos `_ yum install sg3_utils`

```
=====
SAS Drive FW Download - use sg_write_buffer
example loads "ST9600205SS_0002.lod" firmware
=====
```

1. Be in the directory where the .lod files is located then run the `sg_write_buffer` cmd. More details below.
2. On larger code files you must specify `--length=file size`. Use `ls -l` command to get the file size of the code (to use in the `sg_write_buffer` command string

```
# ls -l ST9600205SS_0002_servo-x106.lod
-rw-r--r-- 1 root root 1238016 2011-08-17 10:46 ST9600205SS_0002.lod
```

3. Identify the `/dev/sgx` device. Run the `lsscsi -g` command to see target devices `/dev/sg#`

4. Run `sg_write_buffer` command to the target device. Here is an example of a firmware download to a Seagate Constellation ES.1 model ST9600205SS:

```
# sg_write_buffer -v --in=/var/tmp/ST9600205SS_0002.lod --length=1238016 --mode=5
/dev/sg3
Write buffer cmd: 3b 05 00 00 00 00 12 e4 00 00
```

Or `sg_write_buffer -v --in=<fw file name> -l=<file size> -m=5`

5. Then use Linux SMARTCTL to verify what code is on the disk

```
# smartctl -i /dev/sdc
http://smartmontools.sourceforge.net
Device: SEAGATE ST9600205SS Version: 0002
Serial number: 6XR1xxxx
```

Support

Seagate offers technical support for disk drive installation. If you have any questions related to Seagate products and technologies, feel free to submit your request on our web site. See the web site for a list of world-wide telephone numbers.

Seagate Support: <http://www.seagate.com/support-home/>
Contact Us: <http://www.seagate.com/contacts/>

This software uses open source packages obtained with permission from the relevant parties. For a complete list of open source components, sources and licenses, please see our Linux USB Boot Maker Utility FAQ for additional information.

Linux USB Boot Maker FAQ:

http://support.seagate.com/firmware/usbbootmaker_faq.html

Copyright © 2016 Seagate Technology LLC. All rights reserved.

END USER LICENSE AGREEMENT FOR SEAGATE SOFTWARE

PLEASE READ THIS END USER LICENSE AGREEMENT (“EULA”) CAREFULLY. BY CLICKING “I AGREE” OR TAKING ANY STEP TO DOWNLOAD, SET-UP, INSTALL OR USE ALL OR ANY PORTION OF THIS PRODUCT (INCLUDING, BUT NOT LIMITED TO, THE SOFTWARE AND ASSOCIATED FILES (THE “SOFTWARE”), HARDWARE (“HARDWARE”), DISK (S), CD-ROMS, OR OTHER MEDIA) (COLLECTIVELY, THE “PRODUCT”) YOU AND YOUR COMPANY ACCEPT ALL THE TERMS AND CONDITIONS OF THIS EULA. IF YOU ACQUIRE THIS PRODUCT FOR YOUR COMPANY’S USE, YOU REPRESENT THAT YOU ARE AN AUTHORIZED REPRESENTATIVE WHO HAS THE AUTHORITY TO LEGALLY BIND YOUR COMPANY TO THIS EULA. IF YOU DO NOT AGREE, DO NOT CLICK “I AGREE” AND DO NOT DOWNLOAD, SET-UP, INSTALL OR USE THE SOFTWARE.

1. Ownership. Seagate Technology LLC (“Seagate”) and its suppliers own all right, title, and interest in and to the Software, including all intellectual property rights therein. The Software is licensed, not sold. The structure, organization, and code of the Software are the valuable trade secrets and confidential information of Seagate and its suppliers. The Software is protected by copyright and other intellectual property laws and treaties, including, without limitation, the copyright laws of the United States and other countries. The term “Software” does not refer to or include “Third-Party Software”. “Third-Party Software” means certain software licensed by Seagate from third parties that may be provided with the specific version of Software that you have licensed. The Third-Party Software is generally not governed by the terms set forth below but is subject to different terms and conditions imposed by the licensors of such Third-Party Software. The terms of your use of the Third-Party Software are subject to and governed by the respective license terms, except that this Section 1 and Sections 5 and 6 of this Agreement also govern your use of the Third-Party Software. You may identify and view the relevant licenses and/or notices for such Third-Party Software for the Software you have received pursuant to this EULA at <http://www.seagate.com/www/en-us/support/downloads/>. You agree to comply with the terms and conditions contained in all such Third-Party Software licenses with respect to the

applicable Third-Party Software. Where applicable, the URLs for sites where you may obtain source code for the Third Party Software can be found at <http://www.seagate.com/www/en-us/support/downloads/>.

2. Product License. Subject to your compliance with the terms of this EULA, Seagate grants you a personal, non-exclusive, non-transferable, limited license to install and use one (1) copy of the Software on one (1) device residing on your premises, internally and only for the purposes described in the associated documentation. Use of some third party software included on the CD or other media provided with the Product may be subject to terms and conditions of a separate license agreement; this license agreement may be contained in a “Read Me” file located on the CD or other media that accompanies that Product. The Software includes components that enable you to link to and use certain services provided by third parties (“Third Party Services”). Your use of the Third Party Services is subject to your agreement with the applicable third party service provider. Except as expressly stated herein, this EULA does not grant you any intellectual property rights in the Product. Seagate and its suppliers reserve all rights not expressly granted to you. There are no implied rights.

2.1 Software. You are also permitted to make a single copy of the Software strictly for backup and disaster recovery purposes. You may not alter or modify the Software or create a new installer for the Software. The Software is licensed and distributed by Seagate for use with its storage products only, and may not be used with non-Seagate storage product.

3. Restrictions. You are not licensed to do any of the following:

- a. Create derivative works based on the Product or any part or component thereof, including, but not limited to, the Software;
- b. Reproduce the Product, in whole or in part;
- c. Except as expressly authorized by Section 11 below, sell, assign, license, disclose, or otherwise transfer or make available the Product, in whole or in part, to any third party;
- d. Alter, translate, decompile, or attempt to reverse engineer the Product or any part or component thereof, except and only to the extent that such activity is expressly permitted by applicable law notwithstanding this contractual prohibition;
- e. Use the Product to provide services to third parties;
- f. Take any actions that would cause the Software to become subject to any open source license agreement if it is not already subject to such an agreement; and
- g. Remove or alter any proprietary notices or marks on the Product.

4. Updates. If you receive an update or an upgrade to, or a new version of, any Software

(“Update”) you must possess a valid license to the previous version in order to use the Update. All Updates provided to you shall be subject to the terms and conditions of this EULA. If you receive an Update, you may continue to use the previous version(s) of the Software in your possession, custody or control. Seagate shall have no obligation support the previous versions of the Software upon availability of an Update. Seagate has no obligation to provide support, maintenance, Updates, or modifications under this EULA.

5. NO WARRANTY. THE PRODUCT AND THE THIRD-PARTY SOFTWARE ARE OFFERED ON AN “AS-IS” BASIS AND NO WARRANTY, EITHER EXPRESS OR IMPLIED, IS GIVEN. SEAGATE AND ITS SUPPLIERS EXPRESSLY DISCLAIM ALL WARRANTIES OF ANY KIND, WHETHER STATUTORY, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT. SEAGATE DOES NOT PROVIDE THE THIRD PARTY SERVICES AND MAKES NO WARRANTIES WITH RESPECT TO THE THIRD PARTY SERVICES. YOUR USE OF THE THIRD PARTY SERVICES IS AT YOUR RISK.

6. EXCLUSION OF INCIDENTAL, CONSEQUENTIAL, AND CERTAIN OTHER DAMAGES. TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, IN NO EVENT SHALL SEAGATE OR ITS LICENSORS OR SUPPLIERS BE LIABLE FOR ANY SPECIAL, INCIDENTAL, PUNITIVE, INDIRECT, OR CONSEQUENTIAL DAMAGES WHATSOEVER (INCLUDING, BUT NOT LIMITED TO, DAMAGES FOR LOSS OF PROFITS OR CONFIDENTIAL OR OTHER INFORMATION, FOR BUSINESS INTERRUPTION, FOR PERSONAL INJURY, FOR LOSS OF PRIVACY, FOR FAILURE TO MEET ANY DUTY INCLUDING OF GOOD FAITH OR REASONABLE CARE, FOR NEGLIGENCE, AND FOR ANY OTHER PECUNIARY OR OTHER LOSS WHATSOEVER) ARISING OUT OF OR IN ANY WAY RELATED TO THE USE OF OR INABILITY TO USE THE PRODUCT OR ANY PART OR COMPONENT THEREOF OR RELATED SERVICE OR ANY THIRD PARTY SERVICES, OR OTHERWISE UNDER OR IN CONNECTION WITH ANY PROVISION OF THE EULA, EVEN IN THE EVENT OF THE FAULT, TORT (INCLUDING NEGLIGENCE), MISREPRESENTATION, STRICT LIABILITY, BREACH OF CONTRACT, OR BREACH OF WARRANTY OF SEAGATE OR ITS LICENSORS OR SUPPLIERS, AND EVEN IF SEAGATE OR ITS LICENSOR OR SUPPLIER HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES AND NOTWITHSTANDING ANY FAILURE OF THE ESSENTIAL PURPOSE OF THIS AGREEMENT OR ANY REMEDY.

7. LIMITATION OF LIABILITY. NOTWITHSTANDING ANY DAMAGES THAT YOU MIGHT INCUR FOR ANY REASON WHATSOEVER, THE ENTIRE LIABILITY OF SEAGATE UNDER ANY PROVISION OF THIS EULA AND YOUR EXCLUSIVE REMEDY HEREUNDER SHALL BE LIMITED TO, AND IN NO EVENT WILL SEAGATE'S TOTAL CUMULATIVE DAMAGES EXCEED, THE FEES PAID BY LICENSEE TO SEAGATE FOR THE PRODUCT. ADDITIONALLY, IN NO EVENT SHALL SEAGATE'S LICENSORS OR

SUPPLIERS BE LIABLE FOR ANY DAMAGES OF ANY KIND.

8. Privacy. Seagate's collection, use and disclosure of personally identifiable information in connection with your use of the Product is governed by Seagate's Privacy Policy which is located at <http://www.seagate.com/about/legal-privacy/privacy-statement/>.

9. Indemnification. By accepting the EULA, you agree to indemnify and otherwise hold harmless Seagate, its officers, employees, agents, subsidiaries, affiliates, and other partners from any direct, indirect, incidental, special, consequential or exemplary damages arising out of, relating to, or resulting from your use of the Product or any other matter relating to the Product, including, without limitation, use of any of the Third Party Services.

10. International Trade Compliance. The Software and any related technical data made available for download under this EULA are subject to the customs and export control laws and regulations of the United States ("U.S.") and may also be subject to the customs and export laws and regulations of the country in which the download is contemplated. Further, under U.S. law, the Software and any related technical data made available for download under this EULA may not be sold, leased or otherwise transferred to restricted countries, or used by a restricted end-user (as determined on any one of the U.S. government restricted parties lists, found at <http://www.bis.doc.gov/complianceand enforcement/liststocheck.htm>) or an end-user engaged in activities related to weapons of mass destruction including, without limitation, activities related to designing, developing, producing or using nuclear weapons, materials, or facilities, missiles or supporting missile projects, or chemical or biological weapons. You acknowledge that you are not a citizen, national, or resident of, and are not under control of the governments of Cuba, Iran, North Korea, Sudan or Syria; are not otherwise a restricted end-user as defined by U.S. export control laws; and are not engaged in proliferation activities. Further, you acknowledge that you will not download or otherwise export or re-export the Software or any related technical data directly or indirectly to the above-mentioned countries or to citizens, nationals, or residents of those countries, or to any other restricted end user or for any restricted end-use.

11. General. This EULA between Licensee and Seagate is governed by and construed in accordance with the laws of the State of California without regard to conflict of laws principles. The EULA constitutes the entire agreement between Seagate and you relating to the Product and governs your use of the Product, superseding any prior agreement between you and Seagate relating to the subject matter hereof. If any provision of this EULA is held by a court of competent jurisdiction to be contrary to law, such provision will be changed and interpreted so as to best accomplish the objectives of the original provision to the fullest extent allowed by law and the remaining provisions of the EULA will remain in force and effect. The Product and any related technical data are provided with restricted rights. Use, duplication, or disclosure by the U.S. government is subject to the restrictions as set forth in subparagraph (c)(1)(iii) of DFARS 252.227-7013 (The Rights in Technical Data and Computer Product) or subparagraphs (c)(1) and (2) of 48 CFR 52.227-19 (Commercial Computer Product – Restricted Rights), as applicable.

The manufacturer is Seagate. You may not transfer or assign this EULA or any rights under this EULA, except that you may make a one-time, permanent transfer of this EULA and the Software to another end user, provided that (i) you do not retain any copies of the Software, the Hardware, the media and printed materials, Upgrades (if any), and this EULA, and (ii) prior to the transfer, the end user receiving this EULA and the Software agrees to all the EULA terms. Any attempted assignment in violation of this Section is void. Seagate, the Seagate logo, and other Seagate names and logos are the trademarks of Seagate.

Rev. 09.19.2012

GNU LESSER GENERAL PUBLIC LICENSE

Version 3, 29 June 2007

Copyright © 2007 Free Software Foundation, Inc. <<http://fsf.org/>>

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

This version of the GNU Lesser General Public License incorporates the terms and conditions of version 3 of the GNU General Public License, supplemented by the additional permissions listed below.

0. Additional Definitions.

As used herein, “this License” refers to version 3 of the GNU Lesser General Public License, and the “GNU GPL” refers to version 3 of the GNU General Public License.

“The Library” refers to a covered work governed by this License, other than an Application or a Combined Work as defined below.

An “Application” is any work that makes use of an interface provided by the Library, but which is not otherwise based on the Library. Defining a subclass of a class defined by the Library is deemed a mode of using an interface provided by the Library.

A “Combined Work” is a work produced by combining or linking an Application with the Library. The particular version of the Library with which the Combined Work was made is also called the “Linked Version”.

The “Minimal Corresponding Source” for a Combined Work means the Corresponding Source for the Combined Work, excluding any source code for portions of the Combined Work that, considered in isolation, are based on the Application, and not on the Linked Version.

The “Corresponding Application Code” for a Combined Work means the object code and/or source code for the Application, including any data and utility programs needed for reproducing the Combined Work from the Application, but excluding the System Libraries of the Combined Work.

1. Exception to Section 3 of the GNU GPL.

You may convey a covered work under sections 3 and 4 of this License without being bound by section 3 of the GNU GPL.

2. Conveying Modified Versions.

If you modify a copy of the Library, and, in your modifications, a facility refers to a function or data to be supplied by an Application that uses the facility (other than as an argument passed when the facility is invoked), then you may convey a copy of the modified version:

- a) under this License, provided that you make a good faith effort to ensure that, in the event an Application does not supply the function or data, the facility still operates, and performs whatever part of its purpose remains meaningful, or
- b) under the GNU GPL, with none of the additional permissions of this License applicable to that copy.

3. Object Code Incorporating Material from Library Header Files.

The object code form of an Application may incorporate material from a header file that is part of the Library. You may convey such object code under terms of your choice, provided that, if the incorporated material is not limited to numerical parameters, data structure layouts and accessors, or small macros, inline functions and templates (ten or fewer lines in length), you do both of the following:

- a) Give prominent notice with each copy of the object code that the Library is used in it and that the Library and its use are covered by this License.
- b) Accompany the object code with a copy of the GNU GPL and this license document.

4. Combined Works.

You may convey a Combined Work under terms of your choice that, taken together, effectively do not restrict modification of the portions of the Library contained in the Combined Work and reverse engineering for debugging such modifications, if you also do each of the following:

- a) Give prominent notice with each copy of the Combined Work that the Library is used in it and that the Library and its use are covered by this License.
- b) Accompany the Combined Work with a copy of the GNU GPL and this license document.
- c) For a Combined Work that displays copyright notices during execution, include the copyright notice for the Library among these notices, as well as a reference directing the user to the copies of the GNU GPL and this license document.
- d) Do one of the following:
 - 0) Convey the Minimal Corresponding Source under the terms of this License, and the Corresponding Application Code in a form suitable for, and under terms that permit, the user to recombine or relink the Application with a modified version of the Linked Version to produce a modified Combined Work, in the manner specified by section 6 of the GNU GPL for conveying Corresponding Source.
 - 1) Use a suitable shared library mechanism for linking with the Library. A suitable mechanism is one that (a) uses at run time a copy of the Library already present on the user's computer system, and (b) will operate properly with a modified version of the Library that is interface-compatible with the Linked Version.
- e) Provide Installation Information, but only if you would otherwise be required to provide such information under section 6 of the GNU GPL, and only to the extent that such information is necessary to install and execute a modified version of the Combined Work produced by recombining or relinking the Application with a modified version of the Linked Version. (If you use option 4d0, the Installation Information must accompany the Minimal Corresponding Source and Corresponding Application Code. If you use option 4d1, you must provide the Installation Information in the manner specified by section 6 of the GNU GPL for conveying Corresponding Source.)

5. Combined Libraries.

You may place library facilities that are a work based on the Library side by side in a single library together with other library facilities that are not Applications and are not covered by this License, and convey such a combined library under terms of your choice, if you do both of the following:

- a) Accompany the combined library with a copy of the same work based on the Library, uncombined with any other library facilities, conveyed under the terms of this License.
- b) Give prominent notice with the combined library that part of it is a work based on the Library, and explaining where to find the accompanying uncombined form of the same work.

6. Revised Versions of the GNU Lesser General Public License.

The Free Software Foundation may publish revised and/or new versions of the GNU Lesser General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Library as you received it specifies that a certain numbered version of the GNU Lesser General Public License “or any later version” applies to it, you have the option of following the terms and conditions either of that published version or of any later version published by the Free Software Foundation. If the Library as you received it does not specify a version number of the GNU Lesser General Public License, you may choose any version of the GNU Lesser General Public License ever published by the Free Software Foundation.

If the Library as you received it specifies that a proxy can decide whether future versions of the GNU Lesser General Public License shall apply, that proxy's public statement of acceptance of any version is permanent authorization for you to choose that version for the Library.